



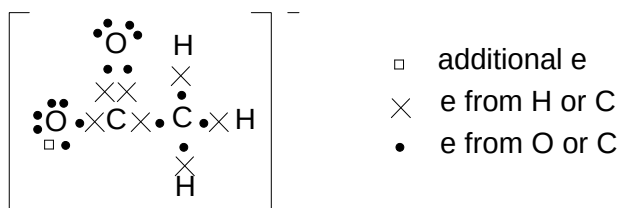
(ii) Down the group, [2]

- ionic radius of M^{2+} increases so the charge density of the M^{2+} decreases
 - M^{2+} becomes less polarising
 - electron cloud of CO_3^{2-} anion is less distorted so the C–O covalent bond within the CO_3^{2-} anions is less weakened down the group.
- Thermal stability of carbonates increases down the group.

[1]: charge density & polarising power of M^{2+} decreases down the group

[1]: extent of distortion & weakening of (C–O) bond decreases and thermal stability increases

(b) (i) [1]



[1]: correct dot-and-cross (ignore missing legend)

(ii) Shape with respect to CH_3CO_2^- : tetrahedral [1]

Shape with respect to CH_3CO_2^- : trigonal planar [1]

(iii) $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$ [1]

(iv) A: $\text{CH}_3\text{CH}_2\text{COCH}_3$ [1]

B: CH_3COCH_3 [1]

(c) Sample C contains manganese(II) ethanoate [3]

Both samples will decompose to form MCO_3 and $(\text{CH}_3)_2\text{CO}$, which gave yellow ppt with 2,4-DNPH.

As $\text{Mn}^{2+}(0.088\text{nm})$ has a smaller ionic radius than $\text{Pb}^{2+}(0.120\text{nm})$, Mn^{2+} ion is able to polarise / distort electron cloud of CO_3^{2-} anion to a greater extent. Hence, MnCO_3 will decompose to give $\text{CO}_2(\text{g})$, which turned limewater cloudy.

[1]: sample C

[1]: quote ionic radius of both M^{2+}

[1]: link to further decomposition of MnCO_3 to give $\text{CO}_2(\text{g})$

2 (a) Co-ordination number is the number of dative bonds attached to the central atom or ion in a complex. [1]

(b) (i) electrons in the 3d and 4s subshells are similar in energy [1] [2]
 thus different number of these electrons are available for use in bond formation / ions formed by using different number of electrons for bonding are of similar stability

[1]: close proximity in terms of energy for 3d and 4s subshell

[1]: different number of e⁻ involved in bonding

(ii) electronic configuration of vanadium in [V(H₂O)₆]²⁺ is [Ar] 3d³ while that in [V(H₂O)₆]³⁺ is [Ar] 3d² [2]

AND

thus [V(H₂O)₆]²⁺ shows a higher relative paramagnetism as it has greater number of unpaired (3d) electrons than [V(H₂O)₆]³⁺.

OR

there are 3 unpaired electrons in the d-orbitals of V²⁺(aq)

and 2 unpaired electrons in the d-orbitals of V³⁺(aq).

Hence the relative paramagnetism of the V²⁺(aq) and V³⁺(aq) ions is 3 : 2.

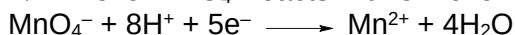
[1]: correct electronic configurations

[1]: compare number of unpaired (3d) electrons and relate to the difference in relative paramagnetism

(c) (i) **Reaction 1**

$$\frac{n(\text{MnO}_4^-)}{n(\text{VO}^{2+})} = \frac{0.0020}{0.010} = \frac{1}{5}$$

⇒ 1 mol of MnO₄⁻ reacts with 5 mol of VO²⁺



⇒ Each mol of MnO₄⁻ gains 5 mol of e⁻

⇒ 5 mol of VO²⁺ loses 5 mol of e⁻

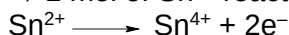
⇒ Each mol of VO²⁺ loses 1 mol of e⁻

∴ final oxidation state of V is (+4 + 1) = +5 [1]

Reaction 2

$$\frac{n(\text{Sn}^{2+})}{n(\text{VO}^{2+})} = \frac{0.0050}{0.010} = \frac{1}{2}$$

⇒ 1 mol of Sn²⁺ reacts with 2 mol of VO²⁺



⇒ Each mol of Sn²⁺ loses 2 mol of e⁻

⇒ 2 mol of VO²⁺ gains 2 mol of e⁻

⇒ Each mol of VO²⁺ gains 1 mol of e⁻

∴ final oxidation state of V is (+4 - 1) = +3 [1]

(ii) Paramagnetism of solution **F** decreases in **Reaction 1** as the number of unpaired (3d) electrons decreases from 1 to 0. [1]

Paramagnetism of solution **F** increases in **Reaction 2** as the number of unpaired (3d) electrons increases from 1 to 2. [1]

(d)

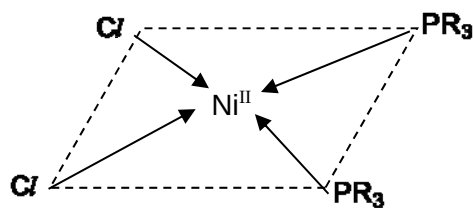
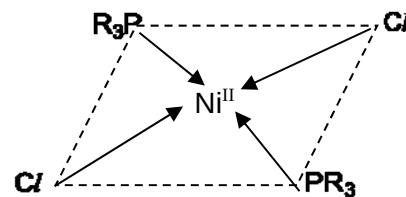
formula of complex	geometry of complex	spin state of Ni ²⁺ in complex
[NiCl ₄] ²⁻	<u>tetrahedral</u>	
[Ni(CN) ₄] ²⁻	<u>square planar</u>	<u>low spin</u>

[2]

[1]: both correct geometry

[1]: correct spin state

(e)

**G****H**

[2]

where R = (-CH₂CH₃) and *lone pair is on P atom***G** has a net dipole moment while **H** does not (dipoles cancel out).(accept **G** in tetrahedral shape)

[1]: correct structure

3 (a) (i) elimination [1]

(ii) It behaves as a Lewis base because it is an electron pair donor / uses its lone pair of electrons on O⁻ to form a dative bond with H⁺. [1]

(iii) [3]

Let the order of reaction with respect to  be x.

Comparing experiments 1 & 2 in which [CH₃CH₂O⁻] is kept constant,

$$\frac{(0.060)^x}{(0.050)^y} = \frac{1.00}{0.833}$$

$$\left(\frac{6}{5}\right)^x = \frac{6}{5}$$

$$x = \underline{1}$$

OR

when  increases 6/5 times, the relative initial rate also decreases 1.00/0.833 ≈ 6/5 times.

Rate of reaction is directly proportional to .

Hence, order of reaction with respect to  is 1.

Let the order of reaction with respect to CH₃CH₂O⁻ be y.

Comparing experiments 2 & 3,

$$\frac{(0.060)^x(0.050)}{(0.050)^x(0.040)} = \frac{0.833}{0.667}$$

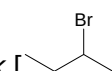
$$\left(\frac{6}{5}\right)^x = 0.999 \approx 1$$

$$x = 0$$

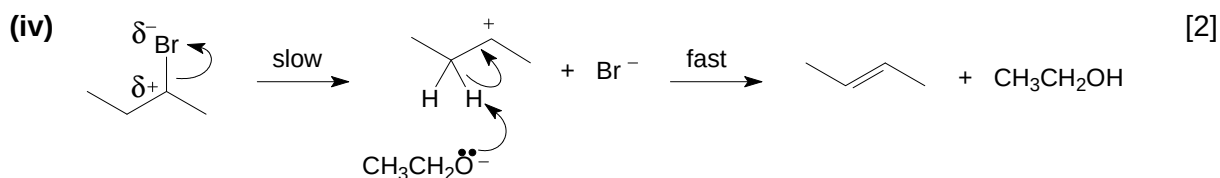
$$\frac{(0.060)^y(0.050)}{(0.050)^y(0.040)} = \frac{0.833}{0.667}$$

$$\left(\frac{6}{5}\right)^y = 0.999 \approx 1$$

$$y = \underline{0}$$

Hence, rate = k 

[1]: order of reaction wrt 
[1]: order of reaction wrt CH₃CH₂O⁻
[1]: rate law



[1]: correct electron pairs movement for both slow and fast step, including showing lone pair on O

[1]: correct intermediate and product generated

- (v) The rate of reaction will be slower when 2-chlorobutane is used instead of 2-bromobutane. [2]

This is because C-Cl bond is stronger than C-Br bond. More energy required to break the C-Cl bond (higher E_a) and reaction is slower.

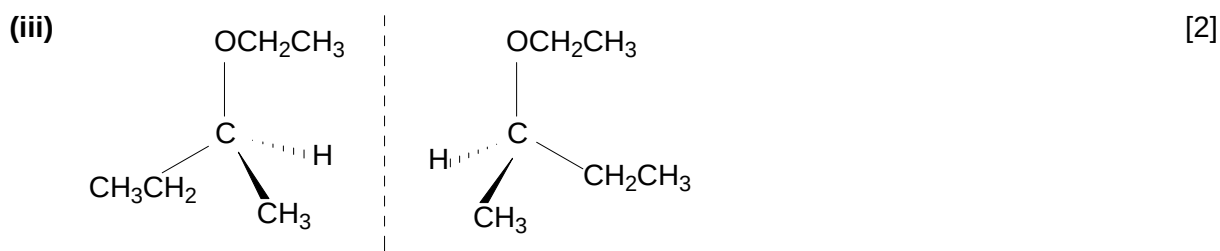
[1]: faster rate

[1]: stronger C-Cl bond

- (b) (i) (unimolecular) nucleophilic substitution, S_N1 [1]

- (ii) The carbocation is trigonal planar wrt the electron deficient C. [1]
 $\text{CH}_3\text{CH}_2\text{O}^-$ acted as a nucleophile and approached the electron-deficient carbon atom from either side with equal probability, producing equimolar mixture of both enantiomers and gives an optically inactive racemic mixture.

[1]: equal probability of attack from either side of planar C^+



[1]: correct by-product

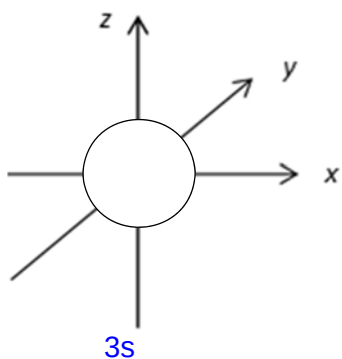
[1]: enantiomers shown correctly

- (iv) The catalyst restricts the nucleophile to attack the electron deficient carbocation from only one side. Hence, one enantiomer to be produced preferentially / no racemic mixture is produced. [1]

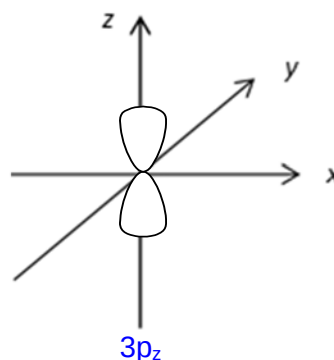
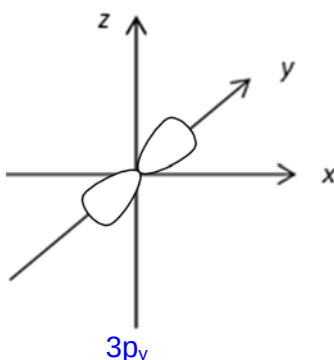
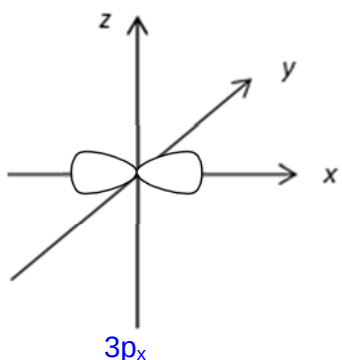
- 4 (a) (i) First ionisation energies decreases down the group. [2]
 Down the group, nuclear charge increases.
 However, valence electron is added to the next electron shell and is further away
and more shielded from nucleus.
 Thus, valence electron experiences weaker electrostatic attraction from the
 nucleus and less energy is required to remove the valence electrons.

[1]: increase in nuclear charge, distance and shielding / screening effect
 [1]: weaker attraction and less energy / decreasing trend

- (ii) electronic configuration of P : $1s^2 2s^2 2p^6 3s^2 3p^3$ [2]



either $3p_x$, $3p_y$ or $3p_z$



[1]: electronic configuration
 [1]: show both 3s and 3p orbitals

- (b) (i) $\text{PH}_3(\text{g}) \longrightarrow \frac{1}{4}\text{P}_4(\text{g}) + \frac{3}{2}\text{H}_2(\text{g})$ [1]

ignore state symbols

- (ii) By Hess Law, [2]

$$\Delta H = 3\text{BE}(\text{P-H}) - \left[\left(\frac{1}{4} \right) (6)\text{BE}(\text{P-P}) + \left(\frac{3}{2} \right) \text{BE}(\text{H-H}) \right]$$

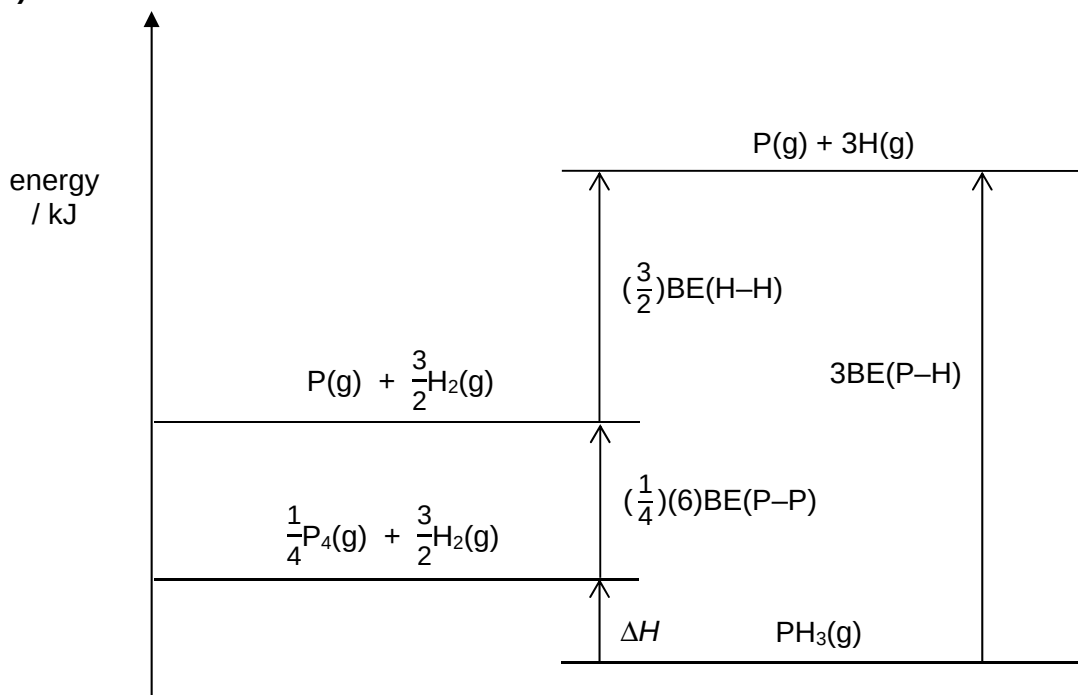
$$= 3(320) - \left[\left(\frac{1}{4} \right) (6)(200) + \left(\frac{3}{2} \right) (436) \right]$$

$$= \underline{+6 \text{ kJ mol}^{-1}}$$

[1]: correct application of Hess' Law
 [1]: correct answers with units

(iii)

[2]



[1]: correct compounds at the 3 other energy levels with corresponding stoichiometry and state symbols

[1]: correct enthalpy terms (or values) and corresponding arrow directions

(iv)

$$k = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{35.0} \\ = \underline{0.0198 \text{ s}^{-1}}$$

[2]

[1]: correct answer

[1]: correct units

(v) When volume of $\text{P}_4(\text{g})$ collected reaches 99% of its expected concentration, 1 % of PH_3 is left. [1]

$$1 \longrightarrow \frac{1}{2} \longrightarrow \frac{1}{4} \longrightarrow \frac{1}{8} \longrightarrow \longrightarrow \frac{1}{100} \left[\left(\frac{1}{2} \right)^n \right]$$

$$\frac{C}{C_0} = \left(\frac{1}{2} \right)^n, \text{ where } n = \text{no. of half-life undergone}$$

$$\frac{1}{100} = \left(\frac{1}{2} \right)^n$$

$$n = \ln \frac{1}{100} \div \ln \frac{1}{2} \\ = 6.64$$

$$\text{estimated time taken for 99 \% of } \text{P}_4 \text{ collected} = 6.64 \times 35.0 \\ = \underline{233 \text{ s}}$$

(accept $6.5 \times 35 = 227.5 \text{ s}$ because $\frac{1}{100}$ is between $\frac{1}{64}$ and $\frac{1}{128}$, i.e. 6th and 7th half-life)

(c) (i) high temperature and low pressure [1]

(ii)
$$n(\text{PH}_3) = \frac{36.0}{31.0 + 3(1.0)}$$
$$= 1.059 \text{ mol}$$
 [2]

$$\begin{aligned} pV &= nRT \\ V &= \frac{nRT}{p} \\ &= \frac{1.059(8.31)(273+327)}{1.34 \times 10^7} \\ &= 3.94 \times 10^{-4} \text{ m}^3 \\ &= \underline{394 \text{ cm}^3} \end{aligned}$$

[1]: correct volume in m^3

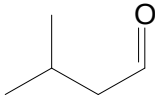
[1]: correct conversion from m^3 to cm^3

(iii) PH_3 behaves non-ideally and volume calculated in (a)(ii) is smaller than the volume of the container. [2]

under high pressure,
the volume of the PH_3 molecules is not negligible / significant. Hence the actual free volume between the gas molecules is smaller.

[1]: recognise that PH_3 is behaving non-ideally and volume is smaller

[1]: correct explanation

- 5 (a) (i) dilute HCl / dilute H₂SO₄ / NaOH(aq) / KOH(aq) (any suitable dilute acid / alkali) [1]
heat under reflux for several hours
- (ii) recommended daily amount of protein = 65 × 1.4 [1]
= 91 g
- (iii) mass of whey protein powder to be consumed = $\frac{91}{84} \times 100$ g [1]
= 108 g
- (accept using per serving = $\frac{91}{21} \times 25$ g)
- (iv) amount of energy contributed by whey protein powder = $\frac{108}{100} \times 1740$ [1]
= 1879 kJ
- amount of energy contributed by other food source = $\frac{80}{20} \times 1879$
= 7516 kJ
- (accept using per serving = $\frac{108}{25} \times 435 \times \frac{80}{20}$ kJ)
- (v) To each of the 2 samples, add K₂Cr₂O₇, dilute H₂SO₄ and heat. [2]
Orange solution remains for phenylamine while orange solution turns green for threonine.
- (accept I₂(aq)/OH⁻ and the corresponding observations – yellow ppt observed for threonine but not phenylamine)
- [1]: correct reagent and condition
[1]: both observations
- (b) (i) Step I: nucleophilic addition [1]
Step IV: hydrolysis [1]
- (ii)  [1]
- (c) (i) When 15.0 cm³ of NaOH(aq) was added, a buffer at its MBC is formed. [1]
pK_{a2} = pH
= 9.60
- K_{a2} = 10^{-9.60}
= 2.5 × 10⁻¹⁰ mol dm⁻³ (shown)
- (ii) **RCH(NH₂)CO₂⁻ + H₂O ⇌ RCH(NH₃⁺)CO₂⁻ + OH⁻** [1]
[OH⁻] > [H⁺] and pH > 7 at point T

- (iii) At 20 cm³ (i.e. 2nd equivalence point), solution contains RCH(NH₂)COO⁻ which is a basic salt [2]

$$K_b = \frac{10^{-14}}{2.5 \times 10^{-10}} = 4.00 \times 10^{-5}$$

$$n(\text{RCH}(\text{NH}_2)\text{CO}_2^-) \text{ formed} = \frac{10}{1000} \times 1.00 = 0.0100 \text{ mol}$$

$$[\text{RCH}(\text{NH}_2)\text{CO}_2^-] \text{ at } 20 \text{ cm}^3 = \frac{0.0100}{(10+20)} \times 1000 = \underline{0.3333 \text{ mol dm}^{-3}}$$

$$K_b = \frac{[\text{OH}^-][\text{RCH}(\text{NH}_3^+)\text{CO}_2^-]}{[\text{RCH}(\text{NH}_2)\text{CO}_2^-]}$$

$$[\text{OH}^-] \approx \sqrt{4.00 \times 10^{-5} \times 0.3333} = 3.65 \times 10^{-3} \text{ mol dm}^{-3}$$

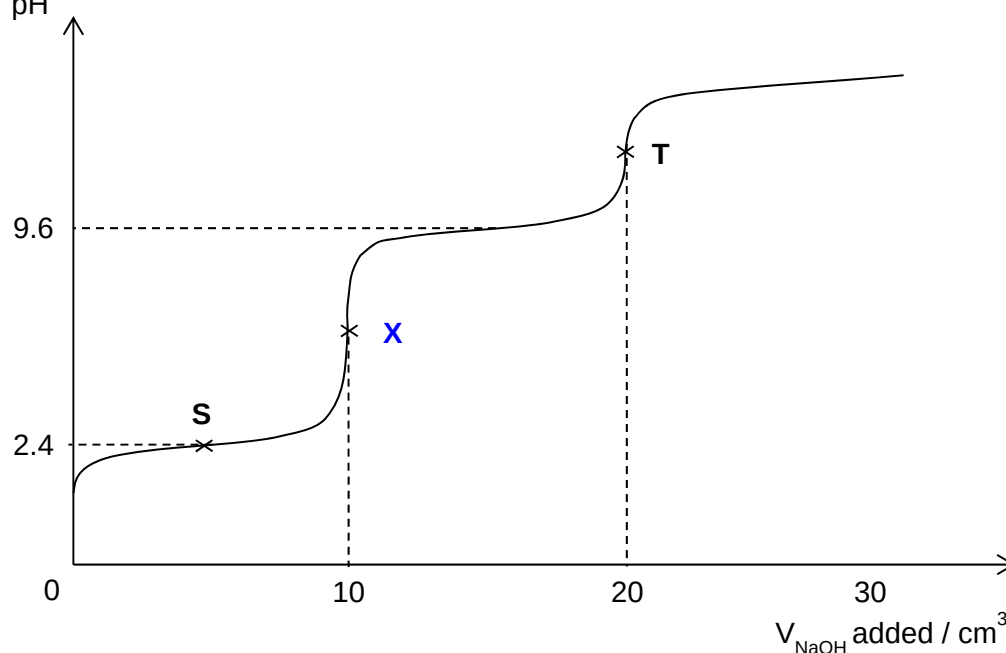
$$\text{pOH} = 2.44$$

$$\text{pH} = \underline{11.6}$$

[1]: [RCH(NH₂)COO⁻]

[1]: pH

- (iv) [1]



[1]: isoelectric point at 10 cm³

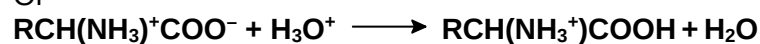
- (v) (CH₃)₂CHCH₂CH(NH₃)⁺CO₂H and (CH₃)₂CHCH₂CH(NH₃)⁺CO₂⁻ [1]

(accept answer in abbreviated form)

[1]: both

- (vi) RCH(NH₃⁺)COO⁻ + H⁺ → RCH(NH₃)⁺COOH [1]

Or



(vii) $[\text{OH}^-] \text{ at } 30 \text{ cm}^3 = \frac{10 \times 1.00}{(10+30)}$ [1]
 $= 0.250 \text{ mol dm}^{-3}$
 $\text{pOH} = 0.60$
 $\text{pH} = \underline{13.4}$

- (d) The $-\text{CO}_2^-$ group formed from dissociation of $\alpha\text{-CO}_2\text{H}$ is closer to the electron withdrawing NH_3^+ group. This disperses the negative charge of the anion to a greater extent and further stabilises the anion. Thus, the ionisation of H^+ is more favoured for the $\alpha\text{-CO}_2\text{H}$. [2]

[1]: closer to EW $-\text{NH}_3^+$

[1]: greater dispersal of negative charge and stabilisation of conjugate base/anion